



AMD Gorgon Point FP8 Engineering Interlock

CUSTOMER ENGINEERING
JULY 2025

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Program Overview ▲



GORGON POINT FAMILY OVERVIEW

- ▲ Gorgon Point 1 APU (Accelerated Processing Unit) is also known as
 - AMD Family 1Ah Models 20h-2Fh Processor
- ▲ Gorgon Point 2 APU (Accelerated Processing Unit) is also known as
 - AMD Family 1Ah Models 60h-63h Processor
- ▲ Gorgon Point 3 APU (Accelerated Processing Unit) is also known as
 - AMD Family 1Ah Models 68h-6Fh Processor
- ▲ Zen5: Core name of the CPU architecture in Gorgon Point FP8 APUs
- ▲ Radeon Navi3.5: Name of the Graphics core in Gorgon Point FP8 APUs
- ▲ Gorgon Point platform includes Hardware Security Processor (HSP) in addition to Platform Security Processor (PSP)
- ▲ Birman+: Gorgon Point FP8 APU-based customer reference board (CRB) based on LPDDR5x with all the required debug and feature test capability
- ▲ Korat+: Gorgon Point FP8 APU-based customer reference board (CRB) based on DDR5 with all the required debug and feature test capability
- ▲ Selkirk: Gorgon Point FP8 APU-based customer reference board (CRB) based on LPCAMM2 with limited debug and feature test capability(support only on Gorgon 1/2)

APU / Platform Overview

CPU CORE & PACKAGE



	Gorgon Point 1/2/3
CPU Architecture	Zen5
CPU Cores	Up to 12 Cores
L1 Cache	L1 Instruction Cache = 32 KB, L1 Data Cache = 48 KB (per core)
L2 Cache	L2 Cache = 1 MB per core
L3 Cache	Up to 24 MB
Graphics	Navi 3.5 graphics with up to 8 WGP (Work Group processors)
Security Processor	Microsoft Security processor to provide fTPM 2.0 and additional security features
Package Type	Lidless BGA, 0.64mm min ball pitch; FP8
Package Size	25mm x 40mm
Max Z-height	1.38 mm
Ball Count	1270

APU & PLATFORM



	Gorgon Point 1/2/3
CPU Architecture	Zen5, Up to 12 Cores
Family	GPT1: Family 1Ah, Models 20h-2Fh GPT2: Family 1Ah, Models 60h-63h GPT3: Family 1Ah, Models 68h-6Fh
Process Technology	TSMC N4
Package / TDP	28W (cTDP 15W-54W)
Primary Power Requirements	Requires 3 Primary Voltages (VDDCR_VDD, VDDCR_SOC and VDDCR_SR)
GPU	Radeon Navi3.5, Up to 8 WGP/16CUs
NPU	GPT1: up to 55 Tops, 32 Byte System Interface GPT2: 50 Tops, 64 Byte System Interface GPT3: 50 Tops, 64 Byte System Interface
Minimum PCB Layers	10 Layers
Reference Board	Birman+ for FP8 LPDDR5x support Korat+ for FP8 DDR5 support Selkirk for FP8 LPCAMM2 support (GPT1/2 only)
dGPU Pairing	Nvidia dGPUs
ROM Size for BIOS	32 MB/64 MB
Software	GorgonPI-FP8, AMD Drivers
S0i3 Modern Standby Support	Yes

FCH & I/O



	Gorgon Point 1/2/3
Memory	LPDDR5x - GPT1/2: Quad 32b Channels, DRAM Down, Support up to LPDDR5x-8533*, LPCAMM2-7500 - GPT3: Quad 32b Channels, DRAM Down, Support up to LPDDR5x-8000 DDR5 Dual 64b Channels, 1 DIMM/Channel, UDIMM, SO-DIMM, DRAM Down, Support up to DDR5-5600
Discrete Graphics Port	PCIe Gen4 Up to 8 lanes and up to 2 ports
GPP	PCIe Gen4 Up to 8 lanes and up to 6 ports
MIPI	No support
Display outputs	5 Ports (eDP, DP, USB4, HDMI, DVI) (3 muxed with USB for USB-C™ with DP Alt. Mode or USB4 or USB3.2) DP2.1 Alt. Mode requires USB-Power Delivery controller separate from SOC
USB4 (40 Gbps)	2 USB4 (40 Gbps)
USB3.2 Gen2 (10Gbps)	1 USB 3.2 Gen2 (10 Gbps) for USB-C™ with DP Alt. Mode port 1 USB 3.2 Gen2 (10 Gbps) for Type A port
USB2.0	3 USB 2.0 Ports (shared controller with USB 3.2 ports)
USB3.2 Gen2 (10Gbps)	1 USB 3.2 Port (dedicated controller) for Secure BIO (* Secure BIO on GPT3 is USB 2.0 Only)
SMBus	2 Ports Shared with I ² C
SPI	Yes
eSPI	Yes
UART	5 Ports: 3 4-wire or 5 2-wire
I ² C / I3C	4 Ports
Clock Generator	Yes
Sensor Fusion Hub	Yes (1 I ² C, 6 GPIOs or 1 I ² C, 1 I3C, 4 GPIOs)
Miracast Wireless Display	Yes

*Refer separate advisory for LPDDR5x 8533 support request.

USB PORT SUPPORT IN DETAILS



Port type	Gorgon Point 1/2/3	Number of Ports
USB-C with DP Alt-mode (C20 PHY)	USB4 40 Gb/s (20 Gb/s per lane) DP2.1 80 Gb/s @ UHBR20 (20 Gb/s per lane) USB3.2 Gen2 10Gb/s	2
USB-C with DP Alt-mode (C10 PHY)	DP2.1 40 Gb/s @ UHBR10 (10 Gb/s per lane) USB3.2 Gen2 10Gb/s	1
USB-A	USB3.2 Gen2 10Gb/s	1
USB2	Shared controller	3
USB3.2	Secure BIO – Dedicated controller (* Secure BIO on GPT3 is USB 2.0 Only)	1

GORGON POINT NOTEBOOK PLATFORMS MEMORY CONFIGURATION

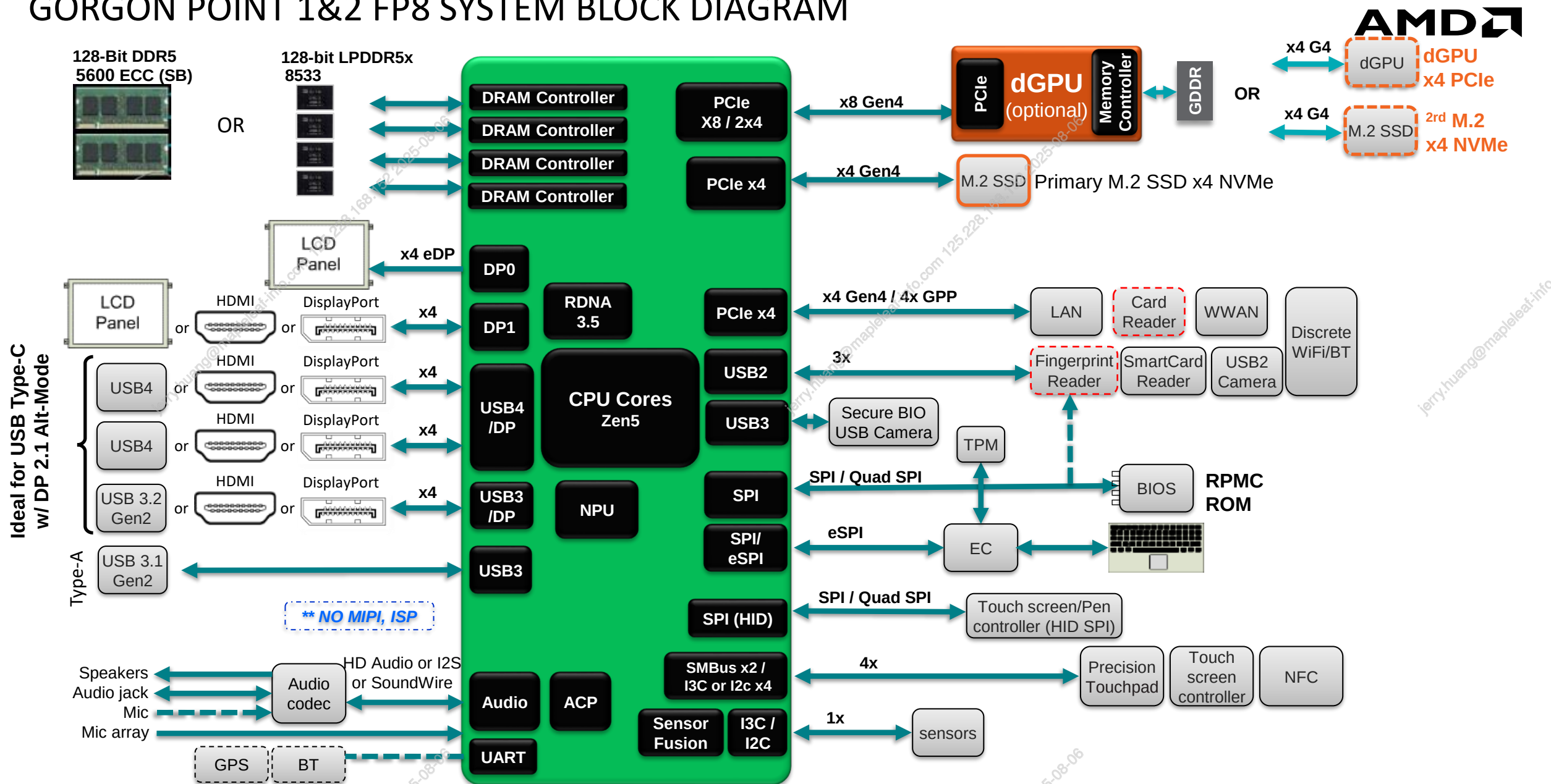


	Gorgon Point 1/2/3 FP8 – LPDDR5	Gorgon Point 1/2/3 FP8 – DDR5
Technology	LPDDR5x 1.05V/0.5V	DDR5 1.1v
Width	128b	128b
Layers	10L with μ Via: GPT1/2 - 8533MT/s*; GPT3 - 8000MT/s 10L with μ Via & 12L without μ Via: 7500MT/s	10L no μ Via
DPC	Memory Down, LPCAMM2**	1DPC
Frequency	GPT1/2 - 8533 MT/s*; GPT3 - 8000MT/s	5600 MT/s
Min Freq	1600 MT/s	2000 MT/s
Error Correction	Link ECC (Option)	ECC (Option)
Form Factor	Memory Down, LPCAMM2 7500**	SODIMM, Memory Down
1up/1down	NA	Yes
DRAM PKG Width	LPDDR5x x32, x64	SODIMM x8, x16
Max Capacity DDR5 @ 16 Gb/die LPDDR5x @ 16 Gb/die	2Rx8 Die, 8DP=64GB (using 4x32 pkg)	SODIMM 1DPC: 48GB/48GB = 96GB max (using 2Rx8) or Memory Down x16: 8GB/8GB = 16GB (using 1Rx16) Memory Down x8 : 48GB/48GB=96GB (using 2Rx8)
Power management	DVFS supported DVFS supported (DVFSQ not supported)	DVFS supported

*Refer separate advisory for LPDDR5x 8533 support request.

** LPCAMM2 is for GPT1/2 only.

GORGON POINT 1&2 FP8 SYSTEM BLOCK DIAGRAM



1. Hardware capability only, for function implementation refer to corresponding guidance.
2. Please refer to supplier firmware readiness and compliance of Components and Peripherals

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Silicon Sample Usage Guidance

GORGON POINT SAMPLE/CUSTOMER TEST ALIGNMENT GUIDANCE



Test Plan Recommendation

▪ **OEM A-Test:**

FP8 - Start test with **GPT1 ES samples and 0.7.0 SW stack**

(GPT1 ES sample for GPT1/2 A phase HW validation purpose)

Functionality: Basic I/O Functionality, Win BOOT, limited thermal calibration, limited OPNs

Usage: Limited Power cycling/GPU stress, Win BOOT, Thermal design, BIOS development, electrical, system bring up & debug

Performance: n/a for Gorgon Point

▪ **OEM B-Test:**

FP8 - Start test with **GPT1/2 PR Samples , GPT3 PC samples and 0.8.0 SW stack**

Functionality: Full I/O Functionality, power cycling, speed & power bin OPNs

Usage: Battery Life testing, Performance Benchmarking, Graphic & CPU Stress Testing, Signal Integrity, Reliability Testing, Regulatory Cert testing, OS Product Feature Testing, Application base Testing, Hardware Config testing

Performance: PC: 80% performance/battery life optimization

▪ **OEM C-Test:**

FP8 - Start Test with **GPT1/2/3 PR samples and 1.0.0 SW stack**

Functionality: No fuse change for silicon (final settings, final GPU drivers, etc.)

Usage: All testing; Regression and & OS Regulatory Certification testing

Performance: PR: 100% performance/battery life optimization

Windows Feature Enablement

Definition:

Enable = feature is working but may not be 100% bug free

Optimized = feature has been optimized and no more changes are being made

GORGON POINT SECURITY FEATURES



System Feature Name	0.7.0	0.8.0	1.0.0	Dependency
AMD Shadow Stack	Enable	Enable	Enable	
Bluetooth Controller Error Recovery	Enable	Enable	Enable	
Enhanced Sign On: MOC (Match On Fingerprint)	Enable	Enable	Enable	
Enhanced Sign On: USB Camera	Enable	Enable	Enable	
Intrusion detection	Enable	Enable	Enable	
Memory Access Protection	Enable	Enable	Enable	
MS Application Guard: Device Credential Guard	Enable	Enable	Enable	
MSFT Windows Hello	Enable	Enable	Enable	
PSP Firmware Anti-Rollback (FAR)	Enable	Enable	Enable	
PlayReady SL3000 Hardware DRM Support	Enable	Enable	Enable	
A/B Partition-based Firmware Update and Recovery	Enable	Enable	Enable	
AMD Secure Parts	Enable	Enable	Enable	
Bitlocker	Enable	Enable	Enable	
Discrete TPM (dTPM)	Enable	Enable	Enable	
Drive Lock	Enable	Enable	Enable	
Microsoft Secure Core PC (SCPC)	Enable	Enable	Enable	
MSFT Windows -System Guard - SMM Protection - SMM Paging Protection	Enable	Enable	Enable	
Platform Secure Boot (PSB)	Enable	Enable	Enable	
Pluton	Enable	Enable	Enable	
Replay protection Monotonic Counter (RPMC) - fTPM Support	Enable	Enable	Enable	
ROM Armor v3	Enable	Enable	Enable	
Virtualization Based Security (VBS)	Enable	Enable	Enable	

GORGON POINT MULTIMEDIA FEATURES



System Feature Name	0.7.0	0.8.0	1.0.0	Dependency
AMD AI-NR Offload on ACP (HiFi5 + NNE110) for PDM/BT/USB	Enable	Enable	Enable	
HDA Offload (Dual LPAP)	Enable	Enable	Enable	
Bluetooth LE Offload (LPAP)	Enable	Enable	Enable	Dependency on 3rd party LE. supported module + Driver. MSFT dependency for LE supported OS
AMD Ultrasound Presence Detection (HDA/Soundwire+ S0)	Enable	Enable	Enable	
Elliptic Ultrasound Presence Detection (HDA/Soundwire+ S0)	Enable	Enable	Enable	Elliptic DSP module.
USB offload (LPAP)	Enable	Enable	Enable	
AOMedia Video 1 (AV1) Codec	Enable	Enable	Enable	
Encode format conversion (tone map, color gamut)	Enable	Enable	Enable	
H.264, H.265 8bit video encode	Enable	Enable	Enable	
Miracast Wireless Display	Enable	Enable	Enable	
RGB reader for encode	Enable	Enable	Enable	
VP9, H.264, H.265 decode with 8bit and 10bit depth	Enable	Enable	Enable	
Bluetooth A2DP offload (LPAP)	Enable	Enable	Enable	Dependency on 3rd party A2DP supported module + Driver
10-bit Display Support	Enable	Enable	Enable	
Soundwire Offload (LPAP)	Enable	Enable	Enable	OEM form factor built with Soundwire peripheral devices
Realtek AI-NR Offload on ACP (HiFi5 + NNE110) for PDM/BT/USB	Enable	Enable	Enable	Dependency on availability of Realtek's NNE based AINR model
Video Processor Engine - VPE	Enable	Enable	Enable	Gorgon Point 1/2 only.

GORGON POINT GRAPHICS & DISPLAY



System Feature Name	0.7.0	0.8.0	1.0.0	Dependency
Mesh Shaders	Enable	Enable	Enable	
Machine Learning Ops	Enable	Enable	Enable	
Ray Tracing Acceleration	Enable	Enable	Enable	
Variable Rate Shading (VRS)	Enable	Enable	Enable	
Display Dynamic Contrast	Enable	Enable	Enable	
Panel Calibration	Enable	Enable	Enable	
DisplayPort 2.0	Enable	Enable	Enable	
DP HBR3 (8.1Gbps)	Enable	Enable	Enable	
eDP Intermediate Link Rate (ILR)	Enable	Enable	Enable	
Display Stream Compression for eDP	Enable	Enable	Enable	
Brightness 3 on SDR panel	Enable	Enable	Enable	
Varibright xLED - Smart Color Profile (SCP)	Enable	Enable	Enable	
Varibright xLED – OLED Edge Diming	Enable	Enable	Enable	
Varibright xLED - Content-Adaptive Contrast Profile (CACP)	Enable	Enable	Enable	
Eyefinity	Enable	Enable	Enable	
Freesync PSR SU (Selective Update)	Enable	Enable	Enable	
FreeSync PSR-SU Rate Control	Enable	Enable	Enable	
FreeSync Premium Pro (PQ HDR)	Enable	Enable	Enable	
HDCP 2.3(4K Netflix)	Disable	Enable	Enable	
HDCP On Demand	Enable	Enable	Enable	
HDMI 2.1 FRL	Enable	Enable	Enable	
Freesync over adapter (DP-to-HDMI Protocol converter)	Enable	Enable	Enable	

GORGON POINT GRAPHICS & DISPLAY CONT.



System Feature Name	0.7.0	0.8.0	1.0.0	Dependency
HDMI 2.1 output over dock adapter and/or protocol converter	Enable	Enable	Enable	
HDMI 2.1 VRR	Enable	Enable	Enable	
High Dynamic Range 10 bit (HDR10)	Enable	Enable	Enable	
Multiple Plane Overlay (MPO)	Enable	Enable	Enable	
Multi-Stream Transport(MST) Docking Stations	Enable	Enable	Enable	
Panel Self Refresh (PSR1)	Enable	Enable	Enable	
Seamless Boot	Enable	Enable	Enable	
Simultaneous Display (up to 4 Displays)	Enable	Enable	Enable	
Dynamic Refresh Rate Control	Enable	Enable	Enable	
ASIC Reset APU	Enable	Enable	Enable	
Varibright – ABM2.4: Ambient light sensor support	Enable	Enable	Enable	a.k.a. Varibright 2.0
PresentAT + Hardware Flip Queue	Enable	Enable	Enable	
Seamless Pixel Rate Switching (SPRS)	Enable	Enable	Enable	
Adaptive Varibright	Enable	Enable	Enable	
Advanced Display Switch (ADS; Win11 Feature)	Enable	Enable	Enable	
VESA Luminance-Based Brightness Control via Aux DPCD	Enable	Enable	Enable	
FreeSync over MST	Enable	Enable	Enable	
FreeSync Replay - support IPX (STX/KRK/GPT) - support Low Hz	Enable	Enable	Enable	LowHz support from 0.8.0

GORGON POINT POWER / PERFORMANCE FEATURES



System Feature Name	0.7.0	0.8.0	1.0.0	Dependency
Advanced Platform Management Link (APML)	Enable	Enable	Enable	
Collaborative Processor Performance Control (CPPC)	Enable	Enable	Enable	
cTDP - Configurable Thermal Design Power	Enable	Enable	Enable	
Dynamic Power Thermal Configurations (DPTC)	Enable	Enable	Enable	
Graphics Power Management	Enable	Enable	Enable	
MS Performance Power Slider Support	Enable	Enable	Enable	
Platform Management Framework (PMF) including Smart PC/Adaptive PC	Enable	Enable	Enable	
Power Delivery (SVI3)	Enable	Enable	Enable	
SDF P-State	Enable	Enable	Enable	
CPU Power Management	Enable	Enable	Enable	
Vari-bright power slider support	Enable	Enable	Enable	
Fan Speed Control	Enable	Enable	Enable	
On-die Temperature sensing Calibration	Enable	Enable	Enable	
Skin Temperature Aware Power Management (STAPM)	Enable	Enable	Enable	
System Temperature Tracking (STT2.1)	Enable	Enable	Enable	
Thermal Throttling	Enable	Enable	Enable	
Thermtrip	Enable	Enable	Enable	
PMF Conferencing Power Mitigation	Enable	Enable	Enable	
Dynamic Voltage and Frequency Scaling (Variable SOC VDD_MEM)	Enable	Enable	Enable	

GORGON POINT SENSOR FEATURE



System Feature Name	0.7.0	0.8.0	1.0.0	Dependency
Accelerometer Sensor	Enable	Enable	Enable	
Ambient Light Sensor (ALS)	Enable	Enable	Enable	
Gyroscope Sensor	Enable	Enable	Enable	
Magnetometer Sensor	Enable	Enable	Enable	
Navigation Compass Sensor	Enable	Enable	Enable	
Proximity Detection/User Presence Sensor	Enable	Enable	Enable	
Secondary Accelerometer Sensor	Enable	Enable	Enable	
Tablet/Book Mode Detection Sensor	Enable	Enable	Enable	

GORGON POINT NPU FEATURES



System Feature	0.7.0	0.8.0	1.0.0	Comment
NPU: Full TOPS Enabled	Enable	Enable	Enable	
NPU: MCDM Driver ¹	Enable	Enable	Enable	¹ Requires Windows 24
Gen AI: Large Language Model (LLM)				Refer to Ryzen AI SW roadmap for LLM list and the optimization details*
Procyon AI benchmark				Refer to Ryzen AI SW roadmap for the enablement and optimization schedule*
Gen AI: Stable Diffusion				Refer to Ryzen AI SW roadmap for the intercept dates*
Windows Next Gen AI				Refer to interlock with MSFT for schedule details

***Please contact your local customer engineering POC for the Ryzen AI roadmap.

GORGON POINT USB FEATURES



System Feature Name	0.7.0	0.8.0	1.0.0	Dependency
Detachable Graphics through USB4	Enable	Enable	Enable	
TBT3.0 compatibility for USB4 ports	Enable	Enable	Enable	
USB 2.0	Enable	Enable	Enable	
USB 3.2 Gen 1 (5Gbps)	Enable	Enable	Enable	
USB 3.2 Gen 2 (10Gbps)	Enable	Enable	Enable	
USB 4.0	Enable	Enable	Enable	
USB Run time D3	Enable	Enable	Enable	

GORGON POINT SYSTEM FEATURES



System Feature Name	0.7.0	0.8.0	1.0.0	Dependency
SD Card 7.0 (External)	Enable	Enable	Enable	
SPI Mode (Dual, Quad) for all 3 SPI ports	Enable	Enable	Enable	
UART (Universal Asynchronous Receiver Transmitter)	Enable	Enable	Enable	
WLAN/WWAN (External)	Enable	Enable	Enable	
AMD Wireless Button Driver	Enable	Enable	Enable	
Cold Boot (S5)	Enable	Enable	Enable	
Hibernate (S4)	Enable	Enable	Enable	
Modern Standby (S0i3)	Enable	Enable	Enable	
eSPI	Enable	Enable	Enable	
I2c	Enable	Enable	Enable	
I3c	Enable	Enable	Enable	Proof of concept
OS Reset	Enable	Enable	Enable	
PCIe (x1, x4, x8) Gen 1 to Gen 4	Enable	Enable	Enable	
Mobile Overclocking (Precision Boost Overdrive)	Enable	Enable	Enable	Consumer Ryzen 9: PBO / PBO Advanced; Consumer Ryzen 7/5: PBO Basic
Mobile Overclocking – DDR5	Enable	Enable	Enable	Memory and Fabric OC for DDR5 / Ryzen 9 Consumer pairings only
Max Supported Memory Speeds - LPDDR5x	Enable	Enable	Enable	Up to 8533 MT/s (GPT1/2) Up to 8000 MT/s (GPT3)
Max Supported Memory Speeds - DDR5	Enable	Enable	Enable	Up to 5600 MT/s

GORGON POINT COMMERCIAL FEATURES



PRO OPN ONLY

System Feature Name	0.7.0	0.8.0	1.0.0	Comment
AMD Integrated Manageability Technology (AIM-T)	Disable	Enable	Enable	Dependent on WLAN Vendor, Single BIOS for multiple LAN.
DASH Profile Support	Disable	Enable	Enable	Dependent on WLAN Vendor, Single BIOS for multiple LAN.
KVM Support	Disable	Enable	Enable	Dependent on WLAN Vendor, Single BIOS for multiple LAN.
Wireless Connectivity (Pre-Boot)	Disable	Enable	Enable	Dependent on WLAN Vendor, Single BIOS for multiple LAN.
AMD Memory Guard	Disable	Enable	Enable	
Memory ECC	Disable	Enable	Enable	PRO OPNs only. LPDDR5 → Link ECC; DDR5 → ECC.
Linux support of 10/12 bit (Deep Color) support for X desktop and OpenGL	Disable	Enable	Enable	Mobile Workstation Feature*
EDID Emulation	Disable	Enable	Enable	Mobile Workstation Feature*
Viewport 2.0	Disable	Enable	Enable	Mobile Workstation Feature*
10-bit Display Support	Disable	Enable	Enable	Mobile Workstation Feature*
Eyefinity	Disable	Enable	Enable	Mobile Workstation Feature*
FRTC Power savings	Disable	Enable	Enable	Mobile Workstation Feature*

*Mobile Workstation Features apply to GPT1 & GPT2 only.

Linux OS Feature Enablement

Definition:

Enable = feature is working but may not be bug free

Optimized = feature has been optimized and no more changes are planned

GORGON POINT 1/2 LINUX FEATURE ENABLEMENT



Feature	Kernel 6.14	Comment
APU GFX (UMA) kernel support	Optimized	
APU GFX (UMA) user-space support	N/A	Mesa 24.1.5 and LLVM 19
USB DP Alt-Mode	Optimized	DP1.4a only
USB 2.0 / 3.2 (5Gbps) / 3.2 (10Gbps)	Enabled	
USB 4.0 CM core functionality	Enabled	
HID Motion Sensors (Accelerometer, Ambient Light, Gyroscope, Magnetometer)	Enabled	
Proximity Detection / Human Presence Detection Sensors	Enabled	
Standby (S3)	Disabled	Not supported
Cold Boot (S5)	Enabled	
S2IDLE (S0i3)	Enabled	
S4 (Hibernate)	Enabled	With condition to disable UEFI Secure Boot, as S4 is Incompatible with UEFI Secure boot
I2C / SMBUS, UART, GPIO, WDT, UCSI	Enabled	
AMD Wireless Button Driver	Enabled	
IOMMU	Enabled	

¹Contact your customer representative regarding status of features prior to these release dates or features not listed.

GORGON POINT 1/2 LINUX FEATURE ENABLEMENT



Feature	Kernel 6.14	Kernel 6.15	Comment
PCIe (x1, x4, x8) Gen 1 to Gen 4	Enabled	Enabled	
DisplayPort 1.4a	Enabled	Enabled	
Embedded Display Port (eDP) Panel	Enabled	Enabled	
4K at 24, 30 and 60 FPS video	Enabled	Enabled	
SPI Mode (Dual, Quad) for all 3 SPI ports	Enabled	Enabled	Only SPI0 (BIOS Flash) support is PoR
AMD Memory Guard (TSME)	Enabled	Enabled	
PSP Anti-rollback	Enabled	Enabled	
PSP FW recovery	Enabled	Enabled	
Secure Boot	Enabled	Enabled	
DASH Manageability (Wired)	N/A	N/A	Target CRB specific discrete NIC only. Support through vendor provided add-on packages.
Audio Coprocessor (ACP) SoundWire	Disabled ¹	Enabled ¹	Only enabled for RT722, RT1320
Audio Coprocessor (ACP) Digital Microphone (DMIC)	Enabled	Enabled	
Platform Management Framework (PMF)	Enabled	Enabled	
Preferred core ranking support	Enabled	Enabled	Requires amd-pstate to be enabled in customer configuration.

¹Contact your customer representative regarding status of features prior to these release dates or features not listed.

GORGON POINT 1/2 LINUX FEATURE ENABLEMENT



Feature	Status	Comment
Neural processing unit (NPU)	Enabled	Kernel support in included in 6.14 Please contact AMD to discuss desired use cases if interested
Z8 Z-state support	Enabled	Not supported with Realtek RTL8111EPP
Panel Self Refresh Selective Update	Enabled	
Freesync Panel Replay	Enabled	
Freesync (Continuously variable refresh rate)	Enabled	Requires appropriate compositor support

GORGON POINT 3 LINUX FEATURE ENABLEMENT



Feature	Kernel 6.17	Comment
APU GFX (UMA) kernel support	Optimized	
APU GFX (UMA) user-space support	N/A	Mesa 25.0.3+ and LLVM 20+
USB DP Alt-Mode	Optimized	DP1.4a only
USB 2.0 / 3.2 (5Gbps) / 3.2 (10Gbps)	Enabled	
USB 4.0 CM core functionality	Enabled	
HID Motion Sensors (Accelerometer, Ambient Light, Gyroscope, Magnetometer)	Enabled	
Proximity Detection / Human Presence Detection Sensors	Enabled	
Standby (S3)	Disabled	Not supported
Cold Boot (S5)	Enabled	
S2IDLE (S0i3)	Enabled	
S4 (Hibernate)	Enabled	With condition to disable UEFI Secure Boot, as S4 is Incompatible with UEFI Secure boot
I2C / SMBUS, UART, GPIO, WDT, UCSI	Enabled	
AMD Wireless Button Driver	Enabled	
IOMMU	Enabled	

¹Contact your customer representative regarding status of features prior to these release dates or features not listed.

GORGON POINT 3 LINUX FEATURE ENABLEMENT



Feature	Kernel 6.17	Comment
PCIe (x1, x4, x8) Gen 1 to Gen 4	Enabled	
DisplayPort 1.4a	Enabled	
Embedded Display Port (eDP) Panel	Enabled	
4K at 24, 30 and 60 FPS video	Enabled	
SPI Mode (Dual, Quad) for all 3 SPI ports	Enabled	Only SPI0 (BIOS Flash) support is PoR
AMD Memory Guard (TSME)	Enabled	
PSP Anti-rollback	Enabled	
PSP FW recovery	Enabled	
Secure Boot	Enabled	
DASH Manageability (Wired)	N/A	Target CRB specific discrete NIC only. Support through vendor provided add-on packages.
Audio Coprocessor (ACP) SoundWire	Enabled ¹	Only enabled for RT722, RT1320
Audio Coprocessor (ACP) Digital Microphone (DMIC)	Enabled	
Platform Management Framework (PMF)	Enabled	
Preferred core ranking support	Enabled	Requires amd-pstate to be enabled in customer configuration.

¹Contact your customer representative regarding status of features prior to these release dates or features not listed.

GORGON POINT 3 LINUX FEATURE ENABLEMENT



Feature	Kernel 6.17	Comment
Neural processing unit (NPU)	Enabled	Kernel support is included in 6.17 Please contact AMD to discuss desired use cases if interested
Z8 Z-state support	Enabled	Not supported with Realtek RTL8111EPP
Panel Self Refresh Selective Update	Enabled	
Freesync Panel Replay	Enabled	
Freesync (Continuously variable refresh rate)	Enabled	Requires appropriate compositor support

Software Enablement

GORGON POINT OPERATING SYSTEMS SUPPORT



Operating System	Alignment	Comments
Windows 11 25H2	- Aligned to production release of drivers - Earlier OS versions planned for interim development releases	64-bit only
Windows 11 24H2/23H2	Support N-1/N-2 OS	64-bit only
Ubuntu Linux	Ubuntu LTS 24.04 with OEM kernel	64-bit only
RHEL Linux	RHEL 10.1	64-bit only

Notes:

- No support Windows7 / Windows 8/ Windows 8.1 /Windows 10
- No support for 32-bit UEFI firmware
- Graphics driver alignment is subject to change
- No support for Legacy BIOS/CSM
- Supported features in Linux always submitted upstream to benefit future community Linux distros (Fedora, Debian)
- Ubuntu and RHEL are only POR with Wayland (Xorg not POR)

SOFTWARE STACK ROADMAP



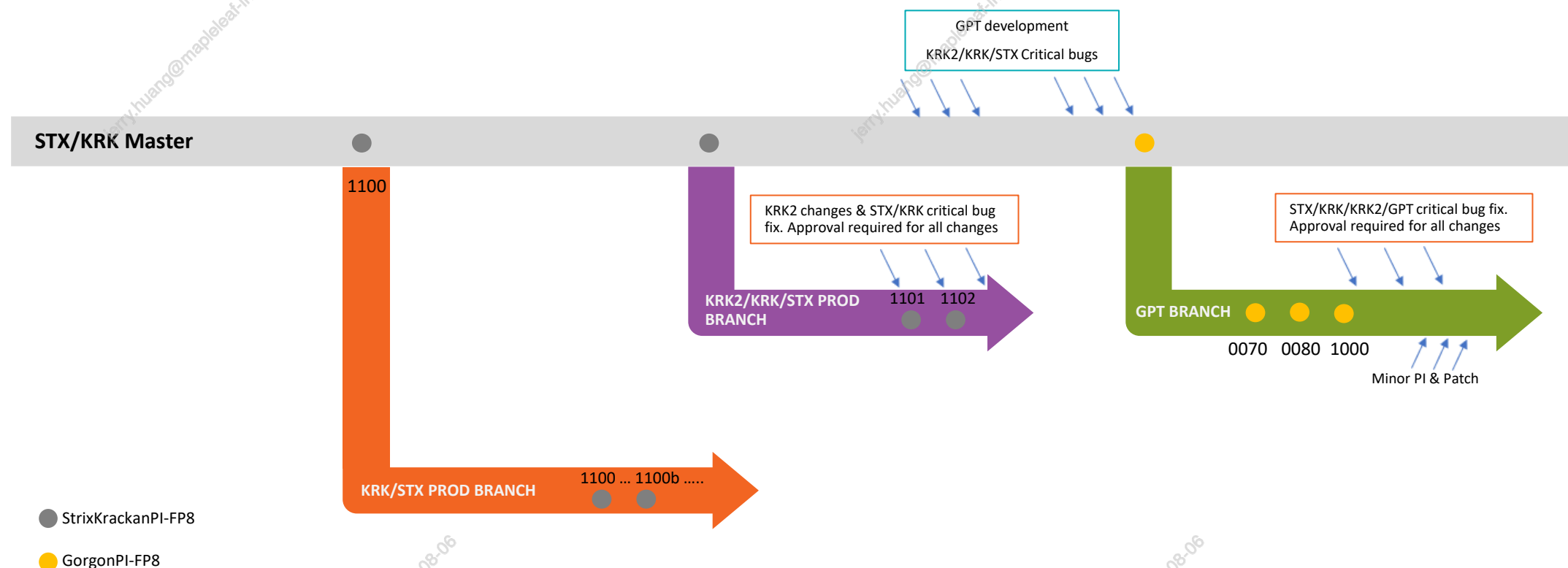
SW Stack	Graphics Driver	AGESA PI	SW Stack Release Date
GPT SW 0.7.0	25.20.00.16	GorgonPI-FP8 0.0.7.0	Jun 09/2025
GPT SW 0.8.0	25.20.xx	GorgonPI-FP8 0.0.8.0	Aug 29/2025
GPT SW 1.0.0*	25.20.xx	GorgonPI-FP8 1.0.0.0	Oct/2025
GPT SW 1.0.1	25.20.xx	GorgonPI-FP8 1.0.0.1	Nov/2025

- Release schedules and driver versions are preliminary and subject to change
- Please always refer to latest release of Gorgon Point Sampling Interlock for schedule with additional granularity when available
- * STX/KRK backwards support from GPT SW 1.0.0.



GORGON PI-FP8 PI STRATEGY

- GorgonPI-FP8 Inherits code from StrixKrackanPI-FP8, and is built from STX/KRK Master codebase.
- Purposes of each PI branches on STX/KRK Master codebase:
 - STX/KRK Master: For development before production branches are available.
 - KRK/STX PROD BRANCH with StrixKrackanPI-FP8: For KRK and STX D5 productions, and 2nd wave of STX LP5 production and post-production.
 - KRK2/KRK/STXPROD BRANCH with StrixKrackanPI-FP8: For KRK2 development and production, and 2nd wave of KRK/STX production and post-production.
 - GORGON BRANCH: Inherits code from StrixKrackanPI-FP8. Includes GPT1/2/3, and STX/KRK/KRK2 backwards supports. STX/KRK backwards support from PI1000.



Graphics, Audio, Display Interfaces ▲

GRAPHICS & VIDEO



	Gorgon Point 1/2/3
Graphics Core	Radeon Navi3.5 Core Architecture Variable Rate Shading (VRS) Ray Tracing Acceleration Mesh Shaders 32b HDR Format (999e5) Filter, Blend, Display ML/Inference Mixed Precision Ops (DLOps) Corner Sampling (PTEx) Lens Matching Rasterization(FSR) Stereo (Double Position) Shader Post depth access Compressed UAV/ROV Shader Access Dirty Rectangle Detect Improved Compute Dispatch PS with only UAV access
DirectX	12
VCN / Video Controller Next	VCN4.0 AOMedia (AV1) encoder & decoder VP9, H.264, H.265 decode with 8bit and 10bit depth and upto 4K@24, 30 and 60 FPS video at Vmin H.264, H.265 8bit video encode up to 4K@30 or 60 fps H.264 B-frame Encode format conversion (tone map, color gamut) RGB reader for encode 8K encode (HEVC) Improved video encoder quality

GRAPHICS & VIDEO



	Gorgon Point 1/2
VPE(Gorgon Point 1/2 only)	Support Video playback 4K60. Input 4K video up to 10bpc (NV12 or P010) or RGB888888 /1010102. Output 4K in RGB format 32bpp (8/10bpc) or 64bpp FP16. Up/down scaling; support H mirroring . 3DLUT tonemapping. Color space conversion.

DISPLAY FEATURE OVERVIEW



Gorgon Point 1/2/3				
Feature		Internal Display (eDP)	External Display - DP/DP alt-mode	External Display - HDMI
FreeSync Dynamic Refresh Rate		Yes	Yes	Yes
Dynamic Contrast		Yes	Yes	Yes
FreeSync Premium Pro – PQ HDR		Yes	Yes	Yes
FreeSync Replay - support IPX (STX/KRK/GPT) - support Low Hz		Yes	N/A	N/A
Seamless Pixel Rate Switching (SPRS)		Yes	N/A	N/A
Varibright – ABM2.4: Ambient light sensor support (LCD panel type)		Yes	N/A	N/A
Varibright xLED (OLED/miniLED)	Smart Color Profile (SCP)	Yes (OLED Only)	N/A	N/A
	OLED Edge Diming	Yes	N/A	N/A
	Content-Adaptive Contrast Profile (CACP)	Yes	N/A	N/A
Adaptive Varibright		Yes	N/A	N/A
Dynamic Refresh Rate Control (DRR; Win11 Feature) v2		Yes	Yes	Yes
Advanced Display Switch (ADS; Win11 Feature)		Yes	N/A	N/A
VESA Luminance-Based Brightness Control via Aux DPCD		Yes	N/A	N/A
Panel Self Refresh	PSR1	Yes	N/A	N/A
	PSR-SU	Yes	N/A	N/A
	PSR-SU Rate Control	Yes	N/A	N/A
High Dynamic Range 10 bit (HDR10)		Yes	Yes	Yes
Brightness 3 on SDR Panel		Yes	N/A	N/A
Panel Calibration		Yes	Yes	Yes
FreeSync over MST		N/A	Yes	N/A

GORGON POINT MOBILE DISPLAY INTERFACES



Interfaces	Gorgon Point 1/2/3
Display Outputs	Five (Two USB-C w/USB4 support; One USB-C w/USB3.2 support; One eDP; One DP/HDMI/eDP) Each 1x4 capable of DP, HDMI or DVI Hot plug support Up to 4 Simultaneous Display
USB4 / DP Alt. Mode over USB-C	Two ports with internal USB4 or USB-C with /DP MUX Requires external PD controller DP2.1 alt mode support HDCP 2.3
DisplayPort	DP 2.1 up to 20G on USB4 Capable ports DP (HBR3, 8.1 Gbps) at 0.6V DSC 1.2a (FEC is required) HDCP 2.3 Multi-Stream Transport (MST) LTTPR
Embedded DisplayPort (eDP)	Version 1.5, HBR3 DSC 1.2a ASSR (required for Content Protection) Dual eDP (DP1 can be used as eDP with backlight support) Intermediate Link Rate (ILR)
HDMI	HDMI 2.1 FRL PHY with DSC, up to 12G with Re-driver DSC1.2a supported in HDMI 2.1 FRL mode HDMI 2.1 VRR for FreeSync using standard HDMI protocol Full HDR support, HDCP 2.3
DVI	1920x1200@60Hz max
Wireless Display	Miracast

HDMI 2.1 FRL & DSC



▲ **HDMI Fixed Rate Link (FRL) enables new bandwidths beyond the 18 Gbps limit of TMDS**

- FRL uses link training and error correction to ensure signal quality (common problems with TMDS)
- FRL enables transport of DSC compressed video to further improve potential video timings

▲ **Gorgon Point FRL enables:**

- 8K60 at Vmax
- 4K120 at Vmin

▲ **DSC enables all Gorgon Point use cases at 6 Gbps FRL rate**

- DSC is optional for displays, so recommend 10 Gbps to ensure interop with displays without DSC

Color Format	4K60	4K120	8K60
SDR 4:4:4 (24bpp)	6 Gbps	8 Gbps	N/S
HDR 4:4:4 (30bpp)	6 Gbps	10 Gbps	N/S
HDR 4:4:4 (36bpp)	6 Gbps	12 Gbps	N/S
HDR 4:4:4 (DSC)	3 Gbps	6 Gbps	6 Gbps
SDR 4:2:0 (12 bpp)	3 Gbps	6 Gbps	8 Gbps
HDR 4:2:0 (15 bpp)	6 Gbps	6 Gbps	10 Gbps
HDR 4:2:0 (18 bpp)	6 Gbps	6 Gbps	12 Gbps

GORGON POINT APU DISPLAY OUTPUT SUPPORT



Maximum Display Connectivity

Port	Version	Example formats Max resolution W/O DSC	Example formats Max resolution With DSC	Max Link Rate	Re-timer/Re-driver
eDP	1.5	3840*2160@120Hz 8bpc 3840*2160@100Hz 8bpc	3840*2160@240Hz 3840*2160@240Hz	HBR3	N
DP	2.1	MST:2 x 3840*2160 @60Hz 10bpc	MST:4 x 3840*2160@60Hz	UHBR20 (On USB4 capable ports)	Not needed if MBDG followed [retimer needed for USB4 ports]
		Single: 3840*2160 @120Hz 10bpc	Single:7680*4320@60Hz		
HDMI	2.1	3840*2160@120Hz 10bpc 8K60A 10bpc (4:2:0)	8K60B(4:4:4)	12G	FRL8 ok for FR-4 up to 5” traces FRL10/12 needs redriver

- Note:
- (1) 4L of HBR3 cannot achieve uncompressed 10bpc (30bpp) for 3840x2160x120; i.e., an uncompressed 10bpc panel could only achieve 3840x2160x96 using HBR3.
 - (2) When running without DSC, we are generally limited by the available link bandwidth rather than SoC constraints.

4K MONITOR DOCKING ON USB-C DISPLAY ALT. MODE



▲ Keeping USB 3.x support to dock active → 2 lane DP Alt Mode

Display Resolution (RGB 4:4:4 format)	4K60 SDR (8bpc)	4K60 HDR ²	4K120 SDR (8bpc)	4K120 HDR ²	APU intercept
USB4 – DP1.4a Tunnelling	1 head ³	1 head ³	N/S ³	N/S ³	HPT/STX/KRK/GPT
USB4 – DP1.4a Tunnelling + DSC ¹	4 heads	4 heads	2 heads	2 heads	HPT/STX/KRK/GPT
DP 2.1 Alt – 20G, 4 lane + USB2	4 heads	4 heads	2 heads	2 heads	STX/KRK/GPT
DP 2.1 Alt – 20G + DSC ¹ , 4 lane + USB2	4 heads	4 heads	4 heads	4 heads	STX/KRK/GPT
DP 2.1 Alt – 20G, 2 lane + USB3	2 heads	2 heads	1 head	1 head	STX/KRK/GPT
DP 2.1 Alt – 20G + DSC ¹ , 2 lane + USB3	4 heads	4 heads	4 heads	4 heads	STX/KRK/GPT

Note¹: Assuming the connected display supports DSC at the specified resolution

Note²: Assuming 8bit with dithering for same B/W as 8bpc

Note³: +1x additional head possible at custom timings without FEC

GORGON POINT AUDIO CO-PROCESSOR SYSTEM FEATURES



Feature Name	Description	Benefits
TinyNNE Addition (SW support of framework)	Added Tiny NNE engine. Support TensorFlow Lite model deployment to Tiny-NNE	Improve battery life on speech and audio inference use cases. Potential lead over Intel, parity with QC/Apple
High performance premium low-power AI noise reduction on Tiny-NNE	Use Tiny-NNE to offload best-in-class AI noise reduction at reduced power. Includes required front-end processing (AEC, BF, AGC as required for Teams cert).	Improved system power savings in Teams with ACP
Improved ACP SDK and tools	Improve ACP SDK and tools for better rapid deployment and reconfiguration for OEMs	Faster time-to-deployment for ACP features
AMD noise reduction and AEC algorithm deployment	Provide a solid, Teams-certified AI-NR and AEC solution to OEMs at no cost	Higher baseline of Teams audio performance and power savings across the ecosystem with AMD

GORGON POINT 1 FP8 DISCRETE GRAPHICS SUPPORT PLAN



28W and APU Pairings	Codename	SKU Name	Enablement Timeline	Modern Standby Support	Smart Access Memory (SAM)	Smart Shift Max	Smart Shift Eco 1.0	CASO	ADS	HG Enabled, Display Mux at dGPU	Device ID	Rev ID
Nvidia 4000/3000 Series	GN21-X2	N/A	0.7.0	Enabled	N/A	N/A	N/A	Enabled	*Note	Enabled		Please see Nvidia for details
	GN20-P0-R											

28W and APU Pairings	Codename	SKU Name	Enablement Timeline	Modern Standby Support	Smart Access Memory (SAM)	Smart Shift Max	Smart Shift Eco 1.0	CASO	ADS	HG Enabled, Display Mux at dGPU	Device ID	Rev ID
Nvidia 5000 Series	GN22-X11 ¹	N/A	TBD	Enabled	N/A	N/A	N/A	Enabled	*Note	Enabled		Please see Nvidia for details
	GN22-X9 ¹											
	GN22-X7 ¹											
	GN22-X6											
	GN22-X4											
	GN22-X2											

*Note - Please refer to MSFT communication
1. Supported only on platforms with <20mm z-height max

GORGON POINT 2 FP8 DISCRETE GRAPHICS SUPPORT PLAN



28W and APU Pairings	Codename	SKU Name	Enablement Timeline	Modern Standby Support	Smart Access Memory (SAM)	Smart Shift Max	Smart Shift Eco 1.0	CASO	ADS	HG Enabled, Display Mux at dGPU	Device ID	Rev ID
Nvidia 4000/3000 Series	GN21-X2	N/A	0.7.0	Enabled	N/A	N/A	N/A	Enabled	*Note	Enabled		Please see Nvidia for details
	GN20-P0-R											

28W and APU Pairings	Codename	SKU Name	Enablement Timeline	Modern Standby Support	Smart Access Memory (SAM)	Smart Shift Max	Smart Shift Eco 1.0	CASO	ADS	HG Enabled, Display Mux at dGPU	Device ID	Rev ID
Nvidia 5000 Series	GN22-X11 ¹	N/A	TBD	Enabled	N/A	N/A	N/A	Enabled	*Note	Enabled		Please see Nvidia for details
	GN22-X9 ¹											
	GN22-X7 ¹											
	GN22-X6											
	GN22-X4											
	GN22-X2											

*Note - Please refer to MSFT communication
1. Supported only on platforms with <20mm z-height max

GORGON POINT 3 FP8 DISCRETE GRAPHICS SUPPORT PLAN



28W and APU Pairings	Codename	SKU Name	Enablement Timeline	Modern Standby Support	Smart Access Memory (SAM)	Smart Shift Max	Smart Shift Eco 1.0	CASO	ADS	HG Enabled, Display Mux at dGPU	Device ID	Rev ID
Nvidia 4000/3000 Series	GN21-X2	N/A	0.7.0	Enabled	N/A	N/A	N/A	Enabled	*Note	Enabled		Please see Nvidia for details
	GN20-P0-R											

28W and APU Pairings	Codename	SKU Name	Enablement Timeline	Modern Standby Support	Smart Access Memory (SAM)	Smart Shift Max	Smart Shift Eco 1.0	CASO	ADS	HG Enabled, Display Mux at dGPU	Device ID	Rev ID
Nvidia 5000 Series	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD

*Note - Please refer to MSFT communication

Memory Implementation Guidance ▲



UNIFIED MEMORY CONTROLLER (UMC)

- ▲ UMC interfaces to the system via Scalable Data Fabric
- ▲ Two UMCs support two DDR5 channels or up to four LPDDR5/x 32-bit channels
 - DDR5: Each UMC controls one memory channel, either 64bit (non-ECC) or 72bit (with ECC)
 - LPDDR5/x: Each UMC controls one memory channel up to two 32bit channels
- ▲ DDR5 Memory Support
 - One SODIMM per Channel or DRAM Down
 - X8 and x16 DRAM Down Support
- ▲ LPDDR5/x Memory support
 - 2x64 or 4 x32 DRAM Down Support
 - Supports x8 or x16 die width
- ▲ Power Management
 - Clock Gating
 - Power Gating
 - 6 Data Fabric P-states
 - Dynamic Voltage and Frequency Scaling (DVFS)
- ▲ UMC Requirements
 - DDR5 or LPDDR5/x support only (no mixed DDR5 or LPDDR5/x)
 - UMC is required to operate at the same MEMCLK frequency, regardless of the channel

VALID MEMORY CONFIGURATIONS AND MAX DATA RATE

AMD

SO-DIMM – DDR5

Memory Type	APU TDP	Board Layers	Number of DIMM Socket Per Channel	Number of DIMMs Populated	DIMM Rank	Max Data Rate
DDR5	All	10	1	2 Max	SR / DR	DDR5-5600

DRAM DOWN – DDR5 OR LPDDR5/x

Memory Type	APU TDP	Board Layers	DRAM Type	Number of DRAM Populated	DIMM Rank	Max Data Rate
DDR5	All	10L without μ Via**	X8 or x16	16 or 8	SRx8 or 16 / DRx8	DDR5-5600
LPDDR5/x	All	10L with μ Via 12L without μ Via**	x32 315 ball (Utilizing x16 or x8 die)	4	SR / DR	LPDDR5x- 7500 (using x16 devices)
		10L with μ Via	x32 315 ball (Utilizing x16 or x8 die)	4	SR / DR	LPDDR5x- 8000 (using x16 devices)
		10L with μ Via	x32 315 ball (Utilizing x16 or x8 die)	4	SR / DR*	LPDDR5x- 8533 (using x16 devices)*
LPDDR5/x	All	10L with μ Via** 12L without μ Via**	x64 496 ball (Utilizing x16 or x8 die)	2	SR / DR	LPDDR5x- 7500 (using x16 devices)

*8533MT/s support on GPT1/2 only. The 8533 MT/s support would be limited to 1RX16 & 2RX16 Memory devices. 2RX8 Memory devices would stay at 8000 MT/s.

**Not validated by AMD CRB.

Two orange geometric shapes: a large parallelogram on the left and a smaller parallelogram on the right, both pointing towards the right.

USB4 Implementation Guidance ▲

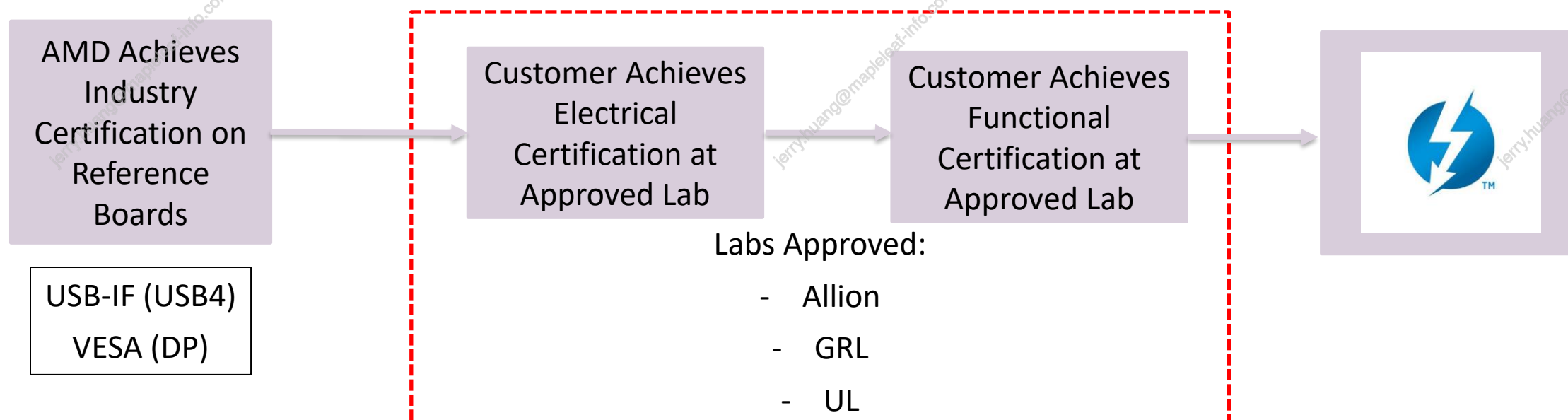
AMD: PATHWAY TO THUNDERBOLT 4 CERTIFICATION



▲ Thunderbolt 4 is a full feature USB4 implementation – AMD systems can qualify for Thunderbolt 4 Branding

▲ Pathway To Thunderbolt 4 Certification and Branding of AMD Gorgon Point Based Customer Platforms:

AMD Customer System



Notes:

1. Requires the customer to submit to 2 locations, where second location does One Day test Only
2. Options for AMD customers: 2 Independent labs or single independent lab company with 2 independent locations
3. Process described in Intel Document # 639816

GORGON POINT: MEETS THUNDERBOLT4 REQUIREMENTS



	USB4 Minimum Requirement	Thunderbolt 4 Requirement	Gorgon Point
Throughput Speed	20Gb/s	40Gb/s	✓
Tunneled DisplayPort	DP 1.1 (4.32Gb/s)	DP 1.4a HBR3 (34.56Gb/s) 2 Independent Streams (SST)	✓
Tunneled PCIE	Optional	PCIE Gen3x4	✓
Thunderbolt 3 Compatibility	Optional	Required	✓
Native USB speed	USB 3.2 G2 (10Gbps)	USB 3.2 (10Gbps)	✓
System Charging	Optional	Required	Platform dependent
Power Delivery for Devices	7.5W Bus Power	15W Bus Power	Platform dependent
USB4 Specification	Compliant	Compliant	✓
DMA Based Protection	Option	Required	✓
USB-PD Revision	PD 3.1	PD 3.1 (Fast Role Swap also recommended)	AVL adherence
External Graphics Support	No	Yes (Windows)	✓
Pre-OS boot From PCIE	No	Recommended	✓

Software Tools

SOFTWARE TOOLS - AMD SPECIALIZED TOOL SUITE



Tool	Purpose
APU Validation Toolkit (AVT)	An integrated toolkit that covers power and thermal management functionality. See more detailed feature list on next slide.
System Stress Test Tool (SST)	Used by validation labs to verify stability of AMD processor-based computing systems • Concurrently runs specialized diagnostics that target system components • Tests may include CPU core, cache, memory, storage USB/eMMC, PCIe, GPU compute, graphics, video, network and other system level test modules. • Used to find problems such as hardware logic errors, bus crosstalk, underpowered signal drivers, and software bugs in device drivers

APU VALIDATION TOOLKIT



Sub Features	Purpose	More Details		
PStates	Used to test performance states on systems using AMD processors	- Core p-state - NB p-state		- Memory p-state - P-state table manipulation
PMM	Allows viewing/configuration of PMM features of processor such as:	- Core Effective frequency - Max-Freq, Min_Freq, OS Power Schemes		- SVI2,SVI3 features - Monitor, ramp up/down CPU utilization - Device States (Windows Only)
CxTest	Tests APU C-State feature	Force processor into various C-States, Inhibit C-states (0% to100%), Monitor C-State residency		
Thermal Charz Tool (TCT)	Thermal testing and characterization	Thermal testing – Runs power virus patterns on user-selected cores		
ThermalChecker	Monitor and configure thermal control functions of systems	SB TSI temperature, PROCHOT status, ACPI Thermal Zones		
Performance Counters	Allows to select and monitor various performance events	Events for – Core, Data Fabric, FCH		
System Access Utility (SAU)	Read/write to various registers in system	- ACPI Tables - MSR (x86) - Memory	- IO/MMIO - PCI - SMBus	- CPUID - Debugger console for – PCI, MSR, IO/MMIO, Memory, SMBus, CPUID - Python Scripting support for above register access.
System Information	Widget displaying System information	- Processor – Name, Family-Model-Stepping, cache size etc.. - BIOS Details, Memory Devices - Size, speed, type, form factor, etc		
Logger	Allows consolidated logging. Can be launched from command line for background logging.			
Cmd line support				

SOFTWARE TOOLS - OTHERS



Tool	Purpose
AMD SystemDeck	Tool to verify SoC Power management feature working for Windows/Linux
AMD Register Explorer (RegXP) -> AMD Register Accessor ARX	Register R/W Debugger
AMDPAT	Platform characterization and bus testing tool used to check memory channels
MEMEYE (1D Support)	DDR Margining Tool for memory margin
AMDPIO	Eye Diagram PCIe/USB Interfaces
BIOS Test Suite (BTS)	Help verify recommended BIOS bit/register settings, combined CPU and GPU
Graphic Diagnostic Suite	APU Graphic diagnostic for board qualification and manufacturing
aHDS	Primary tool for hardware debugging
NetTool	Utility to verify AMD layout guidelines

NDA Documents

GORGON POINT SYSTEM DESIGN DOCUMENTS - [DOCS.AMD.COM](https://docs.amd.com)



Design Guides and Design Files	Part Number	Revision	Release Date
Functional Data Sheet for FP8 Processors	57262	0.93	Nov'24
Electrical Data Sheet for AMD Family 1Ah Models 20h-2Fh Processors	58158	0.80	Nov'24
Electrical Data Sheet for AMD Family 1Ah Models 60h-6Fh Processors	58163	0.90	Mar'25
Motherboard Design Guide for FP8 Processors	57263	1.02	Feb'25
Infrastructure Roadmap for FP8 Processors	57266	1.84	Jun'25
Flotherm Thermal Model User's Guide for FP8 "Krackan" Package Processors	58162	0.70	Feb'24
Flotherm Thermal Model User's Guide for FP8 "Strix" Package Processors	58039	0.90	Feb'25
Thermal Design Guide for FP8 AMD Family 1Ah Models 20h-2Fh Processors	58036	1.02	Feb'25
Thermal Design Guide for AMD Family 1Ah Models 60h-67h Gorgon Point FP8 Processors	58166	0.90	Feb'25
Motherboard Schematic Checklist for FP8 Processors	57264	1.02	Mar'25
Motherboard Layout checklist for FP8 Processors	57265	1.03	Mar'25
Birman+ User Guide for FP8 Processor Platforms	58168	1.01	Feb'24
Korat+ User Guide for FP8 Processor Platform	58502	0.50	Jan'24
Selkirk User Guide for FP8 Processor Platform	68215	1.00	Feb'25
BIOS Porting Guide for Family 1Ah Models 20h-2Fh and 60h-63h FP8 Processors	58224	1.70	Jun'25
Processor Programming Reference (PPR) for AMD Family 1Ah Models 20h-2Fh	57254	1.59	May'24
Processor Programming Reference (PPR) for AMD Family 1Ah Models 60h, Revision A0 Processors	57925	1.50	May'24
AMD BIOS Design Guide for Microsoft Pluton Security Processor Enablement	57174	1.31	Jun'25
AMD Platform Component Requirements	58451	0.50	Feb'24

GORGON POINT SYSTEM DESIGN DOCUMENTS – [DOCS.AMD.COM](https://docs.amd.com)



Design Guides and Design Files	Part Number	Revision	Release Date
AMD Serial VID Interface Version 3 (SVI3) Specification	56413	3.10	Jul'25
AMD Mobile Ergonomic Specifications	57622	1.10	Jul'23
AMD 2-Bits VID Device Reference Guide	57646	0.70	Sept'24
AMD Platform Security Processor BIOS Architecture Design Guide for AMD Client Processor	55758	2.05	Feb'25
AMD Generic Encapsulated Software Architecture (AGESA) V9 Interface Specification	55483	1.72	Apr'23
Secure Biometrics BIOS Implementation Guide	56560	1.01	Jun'23
Approved Vendor List (AVL) for System Components for AMD FP8 Processors	68127	1.20	May'25
AMD FP8 Processor SFH (Sensor Fusion Hub) Design Guideline	57287	0.90	Jan'25
Processor Sensor Fusion Hub (SFH) BIOS Implementation Guide	56371	1.00	Aug'18
Modern Standby BIOS Implementation Guide	56358	1.07	Jun'25
AMD FreeSync™ Specification	56947	4.21	Oct'24
AMD System Peripheral Bus Overview	55861	1.08	Apr'25

Revision History ▲

REVISION HISTORY



Revision	Release Date	Changes
Mar 2025	03/04/2025	• 1st Engineering interlock release
Apr 2025	04/18/2025	• Update family models and NPU • Update sample/customer test alignment guidance • Update feature enablement list • Update SW Stack Roadmap and Gorgon PI Strategy • Update 4K Monitor docking on USB-C Display Alt. Mode table • Update Gorgon Point FP8 graphics support plan • Update system design documents
Jul 2025	07/15/2025	• Update System Block Diagram • Update feature enablement list • Update Operation System Support, SW Stack Roadmap and Gorgon PI Strategy • Update Gorgon Point FP8 graphics support plan • Update system design documents

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